

Validity Analysis: LTZGLUE

Target context: Luxembourgish Professional Newsroom

Overall risk: **HIGH**

Deployment Context

Use case: In a newsroom setting, journalists and editors use LLM-powered software to manage Luxembourgish news articles by applying a model evaluated on headline acceptability, linguistic acceptability, topic classification, and named entity recognition. The system automatically verifies grammatical standards, classifies the article’s domain, and identifies key entities to enable automatic internal linking between related news stories. These outputs determine whether a headline is logically consistent with its content and if the article can be instantly published or must be flagged for manual editorial revision to maintain professional standards.

Target population: Professional journalists, editors, and digital content managers at Luxembourgish media outlets who produce and curate news content in the Luxembourgish language for a local audience.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ✓	3	Moderate gaps	high
Input Content △	2	Significant gaps	medium
Input Form	3	Moderate gaps	medium
Output Ontology △	2	Significant gaps	high
Output Content △	2	Significant gaps	high
Output Form ✓	4	Minor gaps	high
Average	2.7		

△ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark’s test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark’s output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

LTZGLUE is the first GLUE-style benchmark for Luxembourgish and covers all four tasks the newsroom deployment requires (HA, LA, TC, NER), with RTL — the deployment’s primary outlet — as principal data source. However, three of the highest-priority deployment dimensions (IC code-switching, OO output ontology, OC annotator population) show substantial

misalignment. Code-switching, a HIGH-priority operational requirement, is acknowledged as under-represented [Q112] and only the NER comment subcorpus is documented as covering it [Q41]. The LA Ortho ground truth is derived from Spellchecker.lu encoding the formal ZLS standard [Q31, WEB-11], creating systematic divergence from the deployment's flexible journalistic register [A3]. Annotators are student assistants [Q114] with low documented IAA where measured (Cohen's Kappa 0.45 for SA [Q27]) and AI-assisted labelling for RTE [Q63, Q64] and JUDGEWEL [Q37] — none professional journalists. Output form (OF) is well aligned and partially mitigates these concerns. The benchmark itself explicitly cautions against treating its scores as evidence of cultural or demographic coverage [Q120, Q121].

ID	Evidence
Q112	"Coverage across domains, registers, and demographic varieties may also be limited. LTZ displays substantial orthographic and sociolinguistic variation, yet most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts." (p.9)
Q41	"It covers a wider range of text types and registers, including informal and code-mixed writing, and focuses on four primary entity categories (PER, ORG, LOC, GPE)." (p.4)
Q31	"The first class interferes with the subject-verb agreement by changing the conjugated form of the main verb or auxiliary verb. The second class similarly modifies the declined form of the adjective and therefore violates the agreement in case, number, and gender. For the third class, we manipulate the syntax by deleting 2-3 random words from the sentence, depending on the length. The last class impacts the orthography, which is achieved by using data provided by Spellchecker.lu, a semiautomatic spellchecking website frequently used in Luxembourg. We changed one random word in the sentence by using the least frequent variant in the spellchecker data." (p.3)
Q114	"We would like to thank the student assistants for their annotation work." (p.9)
Q27	"We calculated Cohen's Kappa at 0.45." (p.3)
Q63	"To achieve (a), we prompted CHATGPT-5-MINI to judge the texts in the improved data and label their quality as either low, medium, or high, keeping only data rated at least medium, removing nearly 25% of the entire dataset (see Appendix 7.5)." (p.5)
Q64	"For (b), we prompted CHATGPT-5-MINI to verify whether the dataset labels remained correct after the first translation and improvement, outputting true or false for each sentence pair (see Appendix 7.6). Nearly 10% of the labels were false." (p.5)
Q37	"The resulting sentences are then evaluated using LLMs acting as judges, with minimal human verification to calibrate quality thresholds." (p.4)
Q120	"As a result, models may reproduce dominant norms while under-representing regional, sociolectal, or multilingual practices." (p.10)
Q121	"We therefore caution against using benchmark performance as evidence of cultural or demographic coverage." (p.10)
WEB-11	LA Ortho is anchored in Spellchecker.lu which encodes the formal ZLS standard, structurally divorced from the journalistic register the deployment applies (https://data.public.lu/en/organizations/zenter-fir-dletzebuerger-sprooch/)

Practical Guidance

What This Benchmark Measures

LTZGLUE provides a credible first-pass evaluation of generic Luxembourgish NLU capability across the deployment's four required tasks, with strongest signal on output-form compatibility (OF), task-coverage breadth (IO, partially), and an authentic RTL-grounded text source for HA, NER, and TC. Strong encoder performance on TC and NER [Q94, Q105] suggests these tasks are reasonably tractable.

Construct Depth

The benchmark probes surface-level NLU constructs (categorical classification, sequence labelling) but does not probe the deployment's actual ground truth — professional editorial judgment under flexible journalistic register with dense code-switching. LA in particular operationalises formal-orthographic correctness via LOD/Spellchecker [Q30, Q31] rather than journalistic acceptability; HA operationalises TF-IDF structural mismatch [Q22] rather than editorial coherence. Output Content (OC) and Output Ontology (OO) gaps mean high benchmark scores would not directly transfer to editorial-quality judgments.

What Else You Need

To complete the assessment, the deployer should: (a) commission re-annotation of held-out HA and LA samples by professional Luxembourgish editors to address OC and OO gaps; (b) audit benchmark instances for code-switching density

and supplement with a code-switched evaluation set to address the HIGH-priority IC gap; (c) re-evaluate or expand the TC taxonomy against the deploying outlet's actual editorial categories (IO); (d) reinstate GPE/LOC granularity for cross-border NER use cases (OO); (e) calibrate per-task model confidence scores against editorial flagging thresholds in a shadow-deployment phase before live auto-publication.

ID	Evidence
Q94	"Encoder-based models perform strongly across most settings, particularly on structurally complex and label-sensitive tasks, confirming findings from prior work on multilingual and low-resource NLU (Wu and Dredze, 2019; Conneau et al., 2020)." (p.7)
Q105	"First, MMBERT-base consistently achieves the strongest or near-strongest performance across almost all tasks, combining high mean F1 scores with comparatively low variance, suggesting that broad multilingual pre-training with sufficient LTZ exposure yields stable and transferable representations." (p.8)
Q30	"The sentences are derived from the Luxembourgish Online Dictionary (LOD) and are manipulated using the tags available in the dataset." (p.3)
Q31	"The first class interferes with the subject-verb agreement by changing the conjugated form of the main verb or auxiliary verb. The second class similarly modifies the declined form of the adjective and therefore violates the agreement in case, number, and gender. For the third class, we manipulate the syntax by deleting 2-3 random words from the sentence, depending on the length. The last class impacts the orthography, which is achieved by using data provided by Spellchecker.lu, a semiautomatic spellchecking website frequently used in Luxembourg. We changed one random word in the sentence by using the least frequent variant in the spellchecker data." (p.3)
Q22	"The resulting negative examples remain topically related but are temporally and structurally mismatched, forcing models to attend to article content rather than surface cues." (p.3)